MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, ONE TWISTED PAIR AWG-20, FOR SUBMARINE OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE 1PR-A20E

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction

First - Conductor of high strength copper alloy (Phelps-Dodge #PD-135 or equal, silver coated, size AWG-20, class B stranding, (7 by 0.0126-inch wire).

Second - Insulation of ethylene tetrafluoroethylene (TEFZEL), with solid colors (see fourth). Nominal wall thickness 0.010 inch. Maximum overall diameter 0.060 inch.

Third - Insulation jacket of clear natural polyethylene in accordance with L-P-390, type II, class L, grade 3. Nominal wall thickness 0.010 inch. Maximum overall diameter 0.082 inch.

Fourth - Two insulated and jacketed conductors, one white, one red, twisted together with a lay of 1-1/4 ± 1/8 inch.

Fifth - Cable jacket of black polyethylene copolymer, applied so as to completely fill the pair valleys. Unless otherwise specified in the contract or order, the cable jacket material shall be Union Carbide Corp. #DFDA-0588, black #9865 or equal. Cable jacket shall be capable of being removed without incurring damage to the insulation jacket or the insulation. Nominal wall thickness 0.060 inch. Minimum wall thickness at any cross section 0.054 inch. Cable overall diameter shall be 0.285 ± .005 inch.

FSC 6145

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 20°C, maximum.	11.1
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to water.....	2000
Insulation resistance - megohms/1000 feet, minimum.....	5000
Insulation dielectric. Test 100 percent of insulated conductors before application of insulation jacket. Test in accordance with MIL-W-81381, using a test voltage of 8.0 kV (peak), minimum.	
Group A:	
Visual and dimensional.....	No failure
Conductor (before application of insulation)	
Silver coating - thickness in microinches, minimum. (Test in accordance with ASTM B 298)	40
Tensile strength - pounds, minimum.....	55.6
Elongation - percent, minimum..... (Test tensile strength and elongation in accordance with FED-STD-228, method 3212, except as follows. Test not less than 3 specimens per each 1000 feet of conductor. Specimen length shall not be less than 20 inches. Tensile strength shall be in pounds. Specimen zero length shall be as near to 10 inches as practical.)	6.0
Group B:	
Cold bending, cable - at minus 40°C, 3-inch diameter mandrel.....	No damage
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum.....	5000
Elongation - percent, minimum.....	150
Insulation jacket	
Tensile strength - lbf/in^2, minimum.....	1400
Elongation - percent, minimum.....	300
Cable jacket	
Tensile strength - lbf/in^2, minimum.....	1800
Elongation - percent, minimum.....	500

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Insulation	
Air oven (168 hours at 180°C)	
Tensile strength - percent of unaged, minimum	90
Elongation - percent of unaged, minimum	90
Insulation jacket	
Air oven (96 hours at 100 ± 2°C)	
Elongation - percent of original, minimum	60
Cable jacket	
Air oven (96 hours at 100 ± 2°C)	
This test shall be in accordance with MIL-C-915	No failure
Environmental stress crack	
This test shall be performed in accordance with MIL-C-915	No failure
Mutual capacitance - pF/ft at 2 kHz, maximum	25

QUALIFICATION INSPECTION:
Qualification inspection shall include basic electrical, plus all of groups A, B and C.

UNIT ORDERING LENGTH: 1000 feet (nominal).

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-0886-01)

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

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